

$$1) \quad R(k, m) = R(k, m-1) + \frac{R(k, m-1) - R(k-1, m-1)}{4m-1}$$

$$k: n = 2^k$$

$$\therefore n=1; R(0,0) = 11868$$

$$n=2; R(1,0) = 11266$$

$$n=4; R(2,0) = 11133$$

$$n=8; R(3,0) = 11074$$

row \ column

0 11868

1 11266

2 11133

3 11074

$$11065.3333$$

$$11061.7778$$

$$11062$$

$$11060.9333$$

$$11061$$

$$11060.9199$$

$$R(0,1) = R(0,0) + \frac{R(0,0) - R(0,0)}{3}$$

$$= 11266 + \frac{11266 - 11868}{3}$$

$$\Rightarrow 11266 + \frac{(-602)}{3}$$

$$\Rightarrow 11266 - 200.6667$$

$$\approx 11065.33$$

$$R(2,1)$$

$$R(2,1) = R(2,0) + \frac{R(2,0) - R(1,0)}{3}$$

$$= 1113 + \frac{1113 - 11266}{3}$$

$$= 1113 - \frac{153}{3}$$

$$= 1113 - 51$$

$$= 11062$$

$$R(3,1) = R(3,0) + \frac{R(3,0) - R(2,0)}{3}$$

$$= 11074 + \frac{11074 - 1113}{3}$$

$$= 11074 - \frac{39}{3} = 11074 - 13$$

$$= 11061$$

2nd column. $\therefore m=2$. $4^m - 1 \Rightarrow 15$.

$$R(k, 2) = R(k, 1) + \frac{R(k, 1) - R(k-1, 1)}{15}$$

$k=2$

$$R(2, 2) = R(2, 1) + \frac{R(2, 1) - R(1, 1)}{15}$$

$$= 11062 + \frac{11062 - 11065.3333}{15}$$

$$= 11062 + \frac{-3.3333}{15} \approx 11062 - 0.2222$$

$$\approx 11061.7778 \times //$$

$k=3$

$$R(3, 2) = R(3, 1) + \frac{R(3, 1) - R(2, 1)}{15}$$

$$= 11061 + \frac{11061 - 11062}{15} \approx \frac{11061 - 1}{15}$$

$$\approx 11061 - 0.0667$$

$$\approx 11060.9333 \times //$$

3rd column $\therefore m=3$, $4^3 - 1 \Rightarrow 63$.

$$R(3, 3) = R(3, 2) + \frac{R(3, 2) - R(2, 2)}{63}$$

$$= 11060.9333 + \frac{11060.9333 - 11061.7778}{63}$$

$$\approx 11060.9333 - \frac{0.8445}{63}$$

$$\approx 11060.9333 - 0.0134 \Rightarrow 11060.9199 \square //$$